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Dylan Chua

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Education

Harvard University — **Sep 2025 – May 2026**
SM Computational Science and Engineering – GPA: 3.92 / 4.0 — *Massachusetts, United States*

- **Courses**: Stochastic Modelling & Bayesian Inference, High-Performance Computing & Distributed Systems, Advanced Algorithmic Design (MIT), Mathematical Optimisation (MIT), Numerical Methods, Decision Theory

Imperial College London — **Oct 2022 – Jun 2025**
BEng Mechanical Engineering – First Class Honours, 1st in cohort overall — *London, United Kingdom*

- **Awards**: Governors' BEng Prize for Top Scorer Overall, Dean's List, Best Superproject (Design, Make and Test)
- **Relevant Modules**: Mathematics, Computing, Mechatronics & Machine Dynamics

Academic Research & Advanced Projects

AI Research: A Two-Stage Bayesian Framework for LLM Interpretability

- Proposed a novel Bayesian Model Comparison (BMC) framework to rigorously evaluate linear versus circular manifold feature representations in LLMs, using analytic models to formalise geometric intuition.
- Innovated and implemented Sparse Circular Autoencoders (SCAEs) to extract periodic concepts by encoding magnitude and angle, enabling semantic manifold learning when paired with BMC.

Computational Capstone: Collateral Optimisation Engine for Securities Financing

- Headed and architected a containerised, scalable, 4-tier collateral optimisation engine for an external financial client.
- Formulated and solved Mixed-Integer Linear Programming models within the core engine, and designed cryptographic groundwork to guarantee transaction integrity with sealed-bids and certificates of optimality.

Custom GPU Inference Engine for Transformer Architectures

- Developed a bare-metal `C++` inference engine for `Llama-3-8B-Instruct` from scratch, managing memory (`mmap` & GPU) and implementing complete autoregressive decoding without relying on external ML frameworks.
- Engineered optimised `CUDA` kernels for all core operations (`RMSNorm`, `SwiGLU`, `MatMul`) leveraging thread-block shared memory, parallel reductions, and coalesced memory access to optimise arithmetic intensity.

Distributed & Hardware-Accelerated Fast Fourier Transform (FFT)

- Engineered a distributed FFT computation engine, achieving 7× speedup over serial by utilising OpenMP for shared-memory multithreading and MPI for multi-node cluster execution, profiling via Intel VTune.
- Accelerated algorithmic throughput by a further 10× by designing custom CUDA kernels for direct GPU execution, establishing rigorous performance baselines against JIT-compiled JAX arrays.

Numerical Solution of the Black-Scholes Partial Differential Equation (PDE)

- Formulated the mathematical discretisation steps needed to translate the Black-Scholes PDE into a solvable grid
- Utilised both explicit & implicit finite difference methods to robustly evaluate options pricing models in `Python`.

Work Experience

Engineer — **Jun 2026 – Present**
Simulation & Training Systems Hub, Defence Science and Technology Agency — *Singapore*

- Oversee the lifecycle of high-fidelity simulations & training platforms, and drive modern AI adoption in complex systems, by applying computational and system engineering expertise.

Software Development Intern — **Jun 2025 – Aug 2025**
The MathWorks, Inc. — *Cambridge, United Kingdom*

- Designed and shipped a structural decoupling of the AUTOSAR Blockset architecture model from its metamodel, achieving up to a 90% reduction in computational latency and significantly accelerating the end-user experience.

3D Software Development Intern — **Jul 2023 – Sep 2023**
Simulation & Training Systems Hub, Defence Science and Technology Agency — *Singapore*

- Architected a modular Transformer-based 3D semantic segmentation pipeline, automating the end-to-end segmentation of 3D models by mapping categorised 2D texture vertices back to 3D polygons. (Writeup showcase)

Simulations Software Development Intern — **Feb 2022 – Jul 2022**
Command, Control and Communications Development, Defence Science and Technology Agency — *Singapore*

- Developed a high-fidelity drone environment digital twin in `C#`, integrating `ZeroMQ` and `FFmpeg` for low-latency, real-time video telemetry streaming via `RTSP`, as testing infrastructure for computer vision. (Writeup showcase)

Skills

Programming: `Python`, `C++`, `CUDA`, `MATLAB`, `OpenMP`, `MPI`, `C#`, HTML/CSS/React, `TypeScript`, `SQL`, `R`, LaTeX
Systems & Computing: Linux, Git, Docker, `ZeroMQ`, High-Performance Computing (HPC), Distributed Systems
Mathematics & AI: Bayesian Inference, Stochastic Optimisation, Transformer Architectures, Numerical Methods
Languages: English (1st Language), Mandarin, German
Interests: Puzzles & puzzlehunts (MIT Mystery Hunts), Hackathons, Baseball, Football